Tree–network distinguishability fails under the Kimura two-parameter model

Exact counterexamples and a classification of trinets up to level three

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Abstract

Brits, Holtgrefe, van Iersel and Martin (2026) proved that, under the Jukes–Cantor (JC) model of nucleotide substitution, no phylogenetic network containing a non-trivial blob with three or more outgoing edges produces the same leaf-pattern distribution as a phylogenetic tree. Their parameter space consists of stochastic, positive-definite transition matrices with non-trivial reticulation weights. Whether the same holds for the Kimura two-parameter (K2P) model was left open; an earlier version of their paper asserted it, and the assertion was withdrawn. This note shows that the extension is false.

A three-leaf network with two reticulations, whose internal graph is the complete bipartite graph $K_{2,3}$, has exactly the 64-entry leaf-pattern distribution of a three-leaf star tree for an explicit choice of rational network parameters. Every edge matrix of both models comes from an edge-specific continuous-time K2P process in which the transition rate exceeds the transversion rate. On the larger stochastic parameter space, precisely the three unstacked level-two trinets have models intersecting the tree model; only the $K_{2,3}$ seed is certified here under continuous-time, transition-biased parameters. A separate convexity argument excludes its coincidence when both models must share the same prescribed rate ratio greater than one.

The positive cases are explained by an *anchoring* condition, under which the JC induction of Brits et al. remains tied to a single leaf; anchored networks of any level are distinguishable from trees. For networks of level at most three the condition is also necessary: 30 of the 83 reduced level-3 three-leaf networks are anchored, and each of the other 53 coincides with a tree. The coincidences extend to networks with any number of leaves, to every level of at least two, and to the Kimura three-parameter (K3P) model. Every positive and negative claim for the published parameter space rests on a proof or an exact-arithmetic certificate. Two statements about the subset of continuous-time parameters remain conjectural.

1. The question

A phylogenetic network records reticulate events such as hybridisation by allowing a vertex two parents. Under a Markov substitution model on a network, each site evolves along one of the network's displayed trees, chosen by independent coin flips at the reticulations, so the leaf-pattern distribution is a mixture of tree distributions (Gross & Long, 2018; Brits et al., 2026). A basic requirement for inferring reticulation from sequence data is that such mixtures cannot reproduce the distribution of an ordinary tree.

Brits et al. (2026) established this for the JC model. Their argument has three parts. First, restriction to three leaves: if a network with a non-trivial blob produced a tree distribution, so would some three-leaf network (a *trinet*) with a non-trivial three-blob. Second, a *trinet inequality*: the polynomial $q_{ACC}q_{CAC}q_{CCA} - q_{AAA}q_{TCG}^2$ in Fourier coordinates vanishes on star trees and is strictly positive on every trinet with a non-trivial three-blob. Third, the inequality is proved by induction on the number of reticulations, splitting a trinet at a reticulation adjacent to a leaf. The induction needs the polynomial to be symmetric under permutations of the leaves, which holds for JC.

For K2P the same authors proved the level-1 case: the polynomial

$$Q = q_{AGG}\, q_{GAG}\, q_{CCA}^2 - q_{AAA}\, q_{GGA}\, q_{TCG}^2$$

vanishes on star trees and is strictly positive on every three-leaf network with one reticulation adjacent to leaf 3 (their Lemma 4.1; numbering of arXiv version 3). Versions 1 and 2 of the preprint extended this to all levels. Version 3 withdrew the extension, noting that Q is not symmetric under leaf permutations for K2P, so the induction hypothesis need not apply to the smaller networks. The discussion of version 3 lists as open whether the higher-level trinet inequality extends to K2P and K3P.

The answer is negative, and not only because the proof fails: the statement is false. Section 5 gives an exact counterexample; Sections 6–8 determine which trinets of level two and three are distinguishable from trees, and show how the failures propagate to larger networks.

2. Setting

2.1 Networks

We follow the definitions of Brits et al. (2026, Section 2). A *semi-directed* network is obtained from a binary rooted phylogenetic network by suppressing the root and undirecting every edge except the reticulation edges (the two edges entering each reticulation vertex). A *blob* is a maximal connected subgraph without cut edges; an m-blob meets m edges outside itself. A *trinet* is a semi-directed network on three leaves. A *2-sub-blob* is a set W of at least two internal vertices inducing a connected subgraph and joined to the rest of the network by exactly two edges; Brits et al. (2026, Lemma 5.1 and Corollary 5.2) show that suppressing 2-sub-blobs can only enlarge a model, so trinets without 2-sub-blobs (*reduced* trinets) are the cases that matter. A reduced trinet with a non-trivial three-blob consists of that blob and three pendant leaf edges; its level is its number of reticulations.

Deleting one of the two edges entering each reticulation leaves a spanning tree, a *displayed tree*. In the displayed-tree model each reticulation h carries a weight $\delta_h \in (0,1)$ for one of its incoming edges and $1-\delta_h$ for the other, and the network distribution is the weighted mixture of the displayed trees' distributions.

2.2 The K2P model in Fourier coordinates

The root distribution is uniform on the four states. Edge parameters, including transition/transversion ratios, may differ between edges. Outside the stochastic region we use the term *parameterised Fourier model*, not probability distribution.

Identify the nucleotides with the Klein group $\mathbb{Z}_2 \times \mathbb{Z}_2$ by $A=(0,0)$, $C=(0,1)$, $G=(1,0)$, $T=(1,1)$. The transition matrix on an edge e is $M_e(u,v) = f_e(u+v)$, where

$$f(A) = \tfrac{1+2X+Y}{4}, \qquad f(C) = \tfrac{1+Y-2X}{4}, \qquad f(G) = f(T) = \tfrac{1-Y}{4}.$$

Its eigenvalues are $1, X, Y, X$; in the notation of Brits et al. $a^e_A = 1$, $a^e_C = a^e_T = X_e$ and $a^e_G = Y_e$. The substitution class $u \mapsto u + C$ has its own probability and the other two classes share one. Relabelling nucleotides so that this class becomes $A \leftrightarrow G$, $C \leftrightarrow T$ gives Kimura's (1980) model, and we call it the *transition* class. In continuous time, with rate α for transitions and rate β for each of the two transversion classes, an edge of length t has $X = e^{-2(\alpha+\beta)t}$ and $Y = e^{-4\beta t}$.

Three nested parameter regions are used for each edge:

- Θ_0, the space of Brits et al.: the matrix is stochastic, positive definite and not the identity, with $0 < X$, $0 < Y < 1$ and $2X \le 1 + Y$;
- $\Theta_{\mathrm{emb}} \subset \Theta_0$, matrices embeddable in a continuous-time K2P process, which adds $X^2 \le Y$ (that is, $\alpha \ge 0$);
- $\Theta_{\mathrm{tb}} \subset \Theta_{\mathrm{emb}}$, the transition-biased case $X \le Y$ (that is, $\alpha \ge \beta$), the usual situation in real sequence data.

Mixing weights always lie in $(0,1)$. For a network N and a region R, $\mathcal{M}_N(R)$ denotes the set of leaf-pattern distributions obtained from parameters in R. Two networks *coincide* on R if their models intersect, and are *distinguishable* on R if they do not. All positive statements below are proved on the larger box $0 < X, Y < 1$, which needs neither stochasticity nor embeddability.

2.3 Coordinates and the three tree binomials

For a trinet, the Fourier coordinate $q_{x_1x_2x_3}$ is non-zero only when $x_1 + x_2 + x_3 = 0$; there are 16 such coordinates and $q_{AAA} = 1$. The automorphism of the Klein group exchanging C and T fixes every K2P parameter, so q is unchanged when it is applied to all three indices. The remaining 15 coordinates therefore take only nine values. For $\{i,j,k\} = \{1,2,3\}$ write

- A_i for the coordinate with A at leaf i and C at the other two leaves (so $A_3 = q_{CCA}$);

- B_i for the coordinate with A at leaf i and G at the other two (so $B_3 = q_{GGA}$);
- C_i for the coordinate with G at leaf i and C, T at the other two (so $C_3 = q_{TCG}$);

and define the three binomials

$$E_i = A_i^2\, B_j\, B_k - q_{AAA}\, B_i\, C_i^2.$$

The polynomial Q of Brits et al. is E_3. For a star tree whose pendant edge to leaf i has parameters (x_i, y_i), one has $A_i = x_j x_k$, $B_i = y_j y_k$ and $C_i = y_i x_j x_k$, so all three binomials vanish.

3. Two lemmas

Lemma 1 (star-tree test). *Let q be a point of a K2P trinet model with all consistent Fourier coordinates positive (as on Θ_0 or the box; the other 48 coordinates vanish). Then q is the distribution of a K2P star tree with edge parameters in a stochastic region R if and only if $E_1 = E_2 = E_3 = 0$ and the numbers*

$$x_i = \sqrt{A_j A_k / A_i}, \qquad y_i = \sqrt{B_j B_k / B_i}$$

are edge parameters in R. Equivalently, $E_1 = E_2 = E_3 = 0$ holds exactly when there are positive y_1, y_2, y_3 with $B_i = y_j y_k$ and $C_i = y_i A_i$ for each i.

Proof. A star tree has $A_i = x_j x_k$, $B_i = y_j y_k$ and $C_i = y_i x_j x_k$, which give the stated formulas for x_i and y_i. Conversely, with x_i and y_i defined as stated, $A_i = x_j x_k$ and $B_i = y_j y_k$ hold identically, and $E_i = 0$ together with positivity gives $C_i = A_i\sqrt{B_j B_k / B_i} = y_i x_j x_k$. The nine values A_i, B_i, C_i and $q_{AAA} = 1$ determine all 16 Fourier coordinates in both models, by the $C \leftrightarrow T$ symmetry, and the Fourier transform is invertible. □

Lemma 2 (path form). *Let T run over the displayed trees of a trinet, with weight w_T equal to the product of the corresponding mixing weights. In T let c_T be the vertex at which the paths to the three leaves meet, and let X_i^T and Y_i^T be the products of the edge parameters X_e and Y_e along the path from c_T to leaf i. Then*

$$A_i = \sum_T w_T X_j^T X_k^T, \qquad B_i = \sum_T w_T Y_j^T Y_k^T, \qquad C_i = \sum_T w_T\, Y_i^T X_j^T X_k^T.$$

Proof. In a tree, the Fourier parameter used on an edge is indexed by the sum of the states at the leaves on one side of it. Only edges on the three paths separate leaves; on the path to leaf i the index is the state at leaf i, which contributes 1, X_e or Y_e according as it is A, one of C, T, or G. □

Lemma 2 gives a direct reading of the star-tree conditions. Put $\omega_T^{(i)} = w_T X_j^T X_k^T / A_i$; these weights are positive and sum to one, and

$$\frac{C_i}{A_i} = \sum_T \omega_T^{(i)}\, Y_i^T.$$

By Lemma 1, a trinet coincides with a star tree exactly when the G-parameters y_i, which are fixed by the pairwise coordinates B_i alone, can each be written as a weighted mean of the path values Y_i^T, with weights proportional to $w_T X_j^T X_k^T$. Section 9 uses this to explain why JC and K2P behave differently.

4. The six level-2 trinets

We generated every binary rooted phylogenetic network on three labelled leaves with two reticulations (3,276 networks with labelled internal vertices), formed the semi-directed network of each, and kept those whose internal vertices form a single three-blob without 2-sub-blobs. Up to isomorphism there are six such trinets when leaf labels are ignored, and 30 when they are kept. The same procedure with one reticulation returns only the 3-sunlet, and with three reticulations (702,648 rooted networks) it returns 83 trinets up to leaf relabelling. Figure 1 shows the six level-2 trinets, labelled L2-0 to L2-5 as in the accompanying code. Table 1 summarises their structure and the results proved below.

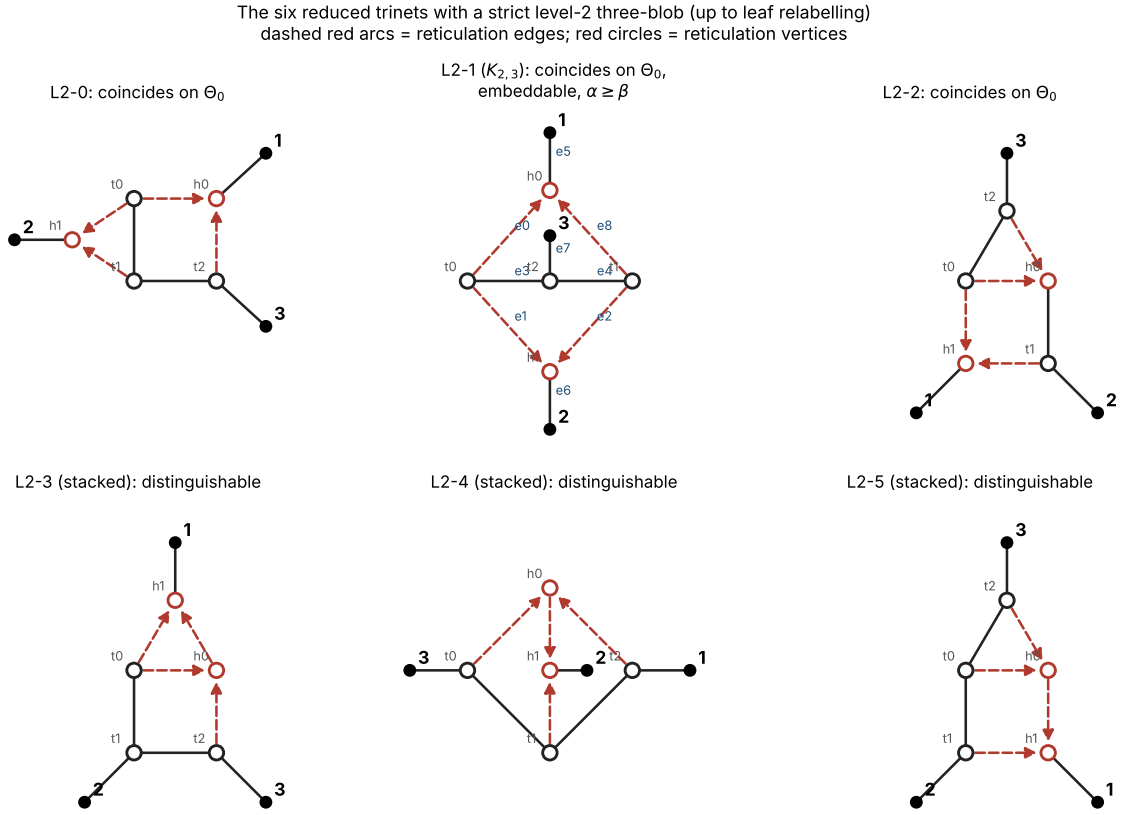


Figure 1: The six reduced trinets with a strict level-2 three-blob, up to relabelling of leaves. Dashed red arcs are reticulation edges and red circles reticulation vertices. Edge names $e_0, \ldots, e_8$ for L2-1 are those of Table 2.

Table 1: Reduced level-two trinets. Coincidence is certified on Θ_0 for the first three; only L2-1 is certified on Θ_{tb}. The two embeddable nonexistence entries remain conjectural. Every internal graph has cycle rank two.

Trinet	Reticulation leaves	Stacked	Anchored	Θ_0	Θ_{emb}
L2-0	1, 2	no	no	yes	open (Conj. 1)
L2-1	1, 2	no	no	yes	yes
L2-2	1	no	no	yes	open (Conj. 1)
L2-3	1	yes	at 1	no	no
L2-4	2	yes	at 2	no	no
L2-5	1	yes	at 1	no	no

In L2-0 the triangle and the 4-cycle share a tree edge; in L2-2 the triangle's reticulation lies on the 4-cycle.

5. An exact counterexample

The trinet L2-1 has internal vertices t_0, t_1, t_2, h_0, h_1. Both reticulations h_0 and h_1 have parents t_0 and t_1; the vertex t_2 is joined to t_0 and t_1; leaves 1, 2 and 3 hang from h_0, h_1 and t_2. Its underlying graph is $K_{2,3}$ with parts $\{t_0, t_1\}$ and $\{t_2, h_0, h_1\}$. Placing the root on the edge t_0t_2 gives a valid rooted network.

Theorem 1. *Give L2-1 the parameters of Table 2, with mixing weight* $1/2$ *at both reticulations. Its K2P leaf-pattern distribution equals that of the star tree whose pendant edges to leaves 1, 2 and 3 have parameters*

$$(x_1, y_1) = \left(\tfrac{\sqrt{105}}{396}, \tfrac{3}{8}\right), \quad (x_2, y_2) = \left(\tfrac{13\sqrt{105}}{900}, \tfrac{3}{16}\right), \quad (x_3, y_3) = \left(\tfrac{\sqrt{105}}{70}, \tfrac{3}{5}\right).$$

Every edge of both models lies in Θ_{tb}*; in particular every edge matrix is stochastic, positive definite, embeddable in continuous time, and has transition rate at least its transversion rate.*

Table 2: Parameters of the counterexample. $\kappa = \alpha/\beta$ is the transition/transversion rate ratio of the corresponding continuous-time edge.

Edge	Endpoints	X_e	Y_e	κ
e_0	$t_0 \to h_0$	31/990	464/1145	6.67
e_1	$t_0 \to h_1$	31/60	53/100	1.08
e_2	$t_1 \to h_1$	1/12	3/20	1.62
e_3	$t_2 — t_0$	51/155	1/2	2.21
e_4	$t_1 — t_2$	1/25	9/10	60.1
e_8	$t_1 \to h_0$	1/2	152/229	2.38
e_5, e_6, e_7	pendant edges	1/2	3/4	3.82

Proof. Choosing e_0 or e_8 at h_0, and e_1 or e_2 at h_1, gives four displayed trees of weight 1/4 each. Leaving aside the pendant edges, their paths from the meeting vertex to leaves 1, 2, 3 are

Displayed tree	Paths to leaves 1, 2, 3	(X_1^T, X_2^T, X_3^T)	(Y_1^T, Y_2^T, Y_3^T)
e_0, e_1 kept	e_0; e_1; e_3	(31/990, 31/60, 51/155)	(464/1145, 53/100, 1/2)
e_0, e_2 kept	e_0e_3; e_2e_4; —	(17/1650, 1/300, 1)	(232/1145, 27/200, 1)
e_8, e_1 kept	e_8e_4; e_1e_3; —	(1/50, 17/100, 1)	(684/1145, 53/200, 1)
e_8, e_2 kept	e_8; e_2; e_4	(1/2, 1/12, 1/25)	(152/229, 3/20, 9/10)

By Lemma 2 the blob contributes

$$(A_1, A_2, A_3) = \left(\tfrac{13}{150}, \tfrac{1}{66}, \tfrac{91}{5940}\right), \quad (B_1, B_2, B_3) = \left(\tfrac{1}{5}, \tfrac{2}{5}, \tfrac{1}{8}\right), \quad (C_1, C_2, C_3) = \left(\tfrac{13}{300}, \tfrac{1}{264}, \tfrac{91}{7425}\right).$$

With $(y_1, y_2, y_3) = (\frac{1}{2}, \frac{1}{4}, \frac{4}{5})$ these satisfy $B_1 = y_2y_3$, $B_2 = y_1y_3$, $B_3 = y_1y_2$ and $C_i = y_iA_i$ for each i, so $E_1 = E_2 = E_3 = 0$. The pendant edges multiply A_i by $\frac{1}{4}$, B_i by $\frac{9}{16}$ and C_i by $\frac{3}{16}$, which leaves the binomials zero. Lemma 1 then gives the star tree; its G-parameters are $\frac{3}{4}y_i$, and $x_i = \frac{1}{2}\sqrt{A_1A_2A_3}/A_i$ with $A_1A_2A_3 = 1183/58806000$, which yields the values stated. The edge conditions are direct: for instance e_1 has $Y - X = 1/75 > 0$, and the tree edge to leaf 2 has $y_2^2 - x_2^2 = 9/256 - 1183/54000 > 0$. □

The 64 probabilities can be checked without relying on the Fourier binomials. Computing each displayed tree's distribution from the transition matrices by summing over all internal states, with exact rational arithmetic, gives for example $p_{AAA} = 1414811/60825600$ and $p_{ACG} = 693553/60825600$, and all 64 entries agree exactly with those of the star tree. The star tree's probabilities are themselves rational, because consistent Fourier-index patterns contain an even number of the letters C, T, so its x-parameters enter only in pairs. Probability-space and Fourier calculations provide separate code paths, not external peer review. The edge-specific rate ratios in Table 2 range from approximately 1.08 to 60.1: transition bias establishes admissibility, not empirical typicality or a homogeneous process.

Three remarks place the example.

It refutes the withdrawn lemma directly. That lemma asserted $E_x > 0$ whenever leaf x lies below a reticulation of a trinet with a non-trivial three-blob. Here $E_1 = E_2 = 0$ at reticulation leaves 1 and 2.

It is not isolated. The Jacobian of (E_1, E_2, E_3) with respect to the 20 network parameters has rank 3 at the point of Table 2, computed exactly. The local coincidence locus is a smooth 17-dimensional submanifold, hence has zero 20-dimensional Lebesgue measure locally. The pendant edges rescale x_i and y_i freely while preserving zero binomials, so the reproduced distributions nevertheless include a non-empty open subset of the six-dimensional star-tree model. These are different ambient spaces. Historical searches sampled 400,000 blob vectors (Y, δ) and found completing X values for approximately 27%, 21% and 15% on the three regions respectively. These are algorithm-dependent completion-search hit rates, not full-network coincidence probabilities or certified volumes. The L2-1 and L2-2 routines scan 400 equally spaced values of t in $[10^{-4}, 1]$ for $X_3 = t\,c(Y_3)$; L2-0 uses an algebraic inequality test. These exploratory searches are not needed for the theorems.

It is a structured mixture, not an arbitrary one. Mixtures can mimic other trees (Matsen & Steel, 2007; Štefankovič & Vigoda, 2007; Matsen et al., 2008). Allman et al. (2012, Theorem 1) exclude mimicry for certain single-tree mixtures with fewer components than states, subject to a topology/refinement qualification; this is not a prohibition on reproducing their own topology. Here four components share edges and parameters as dictated by a network and reproduce the three-leaf star. Kouwenhoven (2025, Chapters 4–5) had already studied failure of a proposed K2P distinguishing polynomial; invariant failure alone does not establish equality of all 64 probabilities. Our certificate checks the latter. Sullivant's graphical-model equivalences concern local network modifications and should not be confused with equality to a tree model.

6. Level two: which trinets coincide with trees

Theorem 2. *On Θ_0, the trinets L2-0, L2-1 and L2-2 each coincide with a star tree, and the trinets L2-3, L2-4 and L2-5 are distinguishable from every trinet without a non-trivial three-blob, star trees included. The distinguishability holds on the whole box $0 < X_e, Y_e < 1$.*

Proof of the coincidences. For L2-1 this is Theorem 1. For L2-0 and L2-2, Appendix A lists all-rational parameters in Θ_0, with mixing weights $(4/5, 1/2)$ and $(2/5, 2/5)$, at which $E_1 = E_2 = E_3 = 0$ and the induced star-tree parameters lie in Θ_0. The exact probability-space calculation again gives 64 identical entries in each case.

Proof of distinguishability. Each of L2-3, L2-4 and L2-5 has a reticulation h adjacent to a leaf x whose two parents are a tree vertex p and another reticulation h'. Deleting the edge $h' \to h$ leaves h' with no descendants, so the displayed network is a star tree and E_x vanishes on it. Deleting the edge $p \to h$ leaves a 3-sunlet whose reticulation h' is adjacent to x. For the 3-sunlet with reticulation leaf x, cycle tree edge f and reticulation edges d, e, a direct calculation gives

$$E_x = \delta(1-\delta)\, P\, X_f^2\, Y_d Y_e\, (1 - Y_f)^2,$$

where P is a product of parameters of the pendant edges, so $E_x > 0$ on the box (this is Lemma 4.1 of Brits et al.). Theorem 3 below, applied with anchor x, completes the proof. A symbolic check confirms each step for all three trinets. □

The certificates for L2-0 and L2-2 use edge matrices outside Θ_{emb}: several edges have $X_e^2 > Y_e$, so these particular network points are not continuous-time embeddable. Their comparison trees are embeddable. Whether all coincidences for these shapes require non-embeddable network edges is unresolved. The reductions in `feasibility_regions.py` and `rational_search_*.py` led to historical completion-search hit rates of approximately 1.7% and 2.2% on Θ_0, and no hits on Θ_{emb}, in 400,000 sampled vectors. Searches with 400 random starts approached boundary points. Neither search proves nonexistence of an interior embeddable coincidence.

Conjecture 1. *L2-0 and L2-2 are distinguishable from star trees on Θ_{emb}.*

The picture on continuous-time parameters is therefore different from that on Θ_0, but the main conclusion does not depend on it: L2-1 coincides with a tree even on Θ_{tb}.

7. Anchored trinets: a sufficient condition at every level

Here is the Fourier action of the suppression used below. A degree-two path replaces (X_e, Y_e) by their coordinatewise products. A parallel pair replaces them by a mixing-weighted coordinatewise mean. For a two-interface sub-blob, condition on its internal reticulation choices and sum out dangling parts. Each displayed tree contributes the product of the Fourier edge parameters along the path between the interfaces, with identity coefficient one. Thus the effective parameters are positive weighted sums of path products, separately for X and Y. Include the two boundary edges in this effective path if necessary; every path is then nonempty and its product is strictly between zero and one. The same calculation applies when the root is trapped inside: uniform-root group symmetry gives a two-interface kernel, and only forward containment in the suppressed model is needed. Products and convex means preserve the positive box and C/T symmetry even when inverse Fourier entries are not nonnegative. This establishes the forward-containment step on that box directly; it does not claim reverse equality there or closure of the continuous-time region under mixtures.

The JC induction of Brits et al. splits a trinet at a reticulation r adjacent to a leaf x. If e' and e'' are the edges entering r and δ the weight of e', the displayed networks $N' = N - e''$ and $N'' = N - e'$ satisfy

$$q = \delta\, q' + (1-\delta)\, q'',$$

and the coordinates with state A at leaf x (here A_x and B_x) are the same in q, q' and q'' (Brits et al., Lemma 2.6). Expanding gives

$$E_x(q) = \delta^2 E_x(q') + (1-\delta)^2 E_x(q'') + \delta(1-\delta)\, K, \qquad K = A_x^2(B'_j B''_k + B''_j B'_k) - 2B_x C'_x C''_x.$$

If $E_x(q') \geq 0$ and $E_x(q'') \geq 0$, the inequality between arithmetic and geometric means gives $K \geq 2A_x^2\sqrt{B'_j B'_k B''_j B''_k} - 2B_x C'_x C''_x \geq 0$. Everything turns on knowing $E_x \geq 0$ for the displayed networks, *at the same leaf* x. In those networks x may no longer lie below a reticulation, and Section 5 shows that E_x can then be zero or negative.

Definition. A reduced trinet N is *anchored at leaf* x if either N has no non-trivial three-blob, or x is adjacent to a reticulation r of N and, for both edges e entering r, the displayed network $N - e$ (with dangling parts removed, vertices of degree two and parallel edges suppressed, and 2-sub-blobs suppressed) is anchored at x.

Theorem 3. *If a trinet N is anchored at x, then $E_x \geq 0$ on the K2P model of N over the box $0 < X_e, Y_e < 1$, and $E_x > 0$ if N has a non-trivial three-blob. In that case N is distinguishable from every trinet without a non-trivial three-blob, on the box and hence on Θ_0.*

Proof. We argue by induction on the number of reticulations. Suppressing 2-sub-blobs, degree-two vertices and parallel edges does not shrink a model (Brits et al., Lemma 5.1; for parallel edges and degree-two vertices the box is closed under products and convex combinations), so it suffices to treat reduced N. If N has no non-trivial three-blob it reduces to a star tree, on which $E_x \equiv 0$. Otherwise the expansion above applies at the reticulation adjacent to x. The points q' and q'' lie in the models of the reduced displayed networks, which are anchored at x and have fewer reticulations, so $E_x(q'), E_x(q'') \geq 0$ by induction, $K \geq 0$, and $E_x(q) \geq 0$. For strictness: suppressing 2-sub-blobs does not destroy a non-trivial three-blob. A 2-sub-blob W inside the blob contains at most one of

the three vertices at which the blob meets its outgoing edges: two of them would contribute two leaving edges, and two more would join W to the rest of the two-edge-connected blob. If one displayed network has a non-trivial three-blob, its E_x is strictly positive by induction and enters with a positive coefficient. If neither does, N has level one by Lemma 5.4 of Brits et al., so it is a 3-sunlet with reticulation leaf x, and $E_x > 0$ by the formula in Section 6. Finally, $E_x \equiv 0$ on every trinet without a non-trivial three-blob, by Lemma 5.1 of Brits et al. and Lemma 1. □

The 3-sunlet is anchored at its reticulation leaf, and so are L2-3, L2-4 and L2-5 at the leaf below their lower reticulation. None of L2-0, L2-1, L2-2 is anchored at any leaf: in each, every leaf below a reticulation has a displayed network that is a 3-sunlet whose reticulation sits above a *different* leaf. That is precisely the configuration in which the withdrawn induction step fails.

8. Propagation, higher levels and more leaves

Two operations carry a coincidence from small networks to larger ones.

Lemma 4 (pendant subtrees). *If a trinet N and a star tree S have the same distribution, attach rooted binary trees T_1, T_2, T_3 at the three leaves of both, with the same parameters. The resulting n-leaf network and n-leaf tree have the same distribution.*

Proof. In every displayed tree, the states at the leaves of T_i depend on the rest only through the state at the root of T_i, with the same conditional law in both models. Summing over the three root states expresses both n-leaf distributions by the same formula in terms of the common trinet distribution. □

Lemma 5 (one more reticulation). *Let N_0 be a reduced trinet that coincides with a star tree at a parameter point θ_0 lying strictly inside a region $R \in \{\Theta_0, \Theta_{\mathrm{emb}}, \Theta_{\mathrm{tb}}\}$, with strictly admissible star-tree parameters, at which the Jacobian of (E_1, E_2, E_3) has rank 3. Let N be any trinet with a reticulation edge c such that $N - c$, after removing dangling parts and suppressing vertices of degree two and parallel edges (but no 2-sub-blob), is isomorphic to N_0. Then N coincides with a star tree at a point with the same three properties.*

Proof. For an effective edge with interior parameters (X, Y), choose extra product factors strictly inside R and sufficiently close to $(1, 1)$, and divide (X, Y) coordinatewise by their product to obtain the remaining factor. Openness of strict admissibility keeps that factor inside R. Such near-identity factors exist in all three regions, for example along a continuous-time path with rates $\alpha > \beta > 0$. The derivatives with respect to the remaining X and Y factors are the nonzero products of the extra factors, so this product map has a local section and is a submersion. For parallel edges use the same interior pair (X, Y) on both edges, with a fixed mixing weight in $(0, 1)$. Their convex combination equals (X, Y), and varying one pair has nonzero scalar derivative in both coordinates. No global convexity of the embeddable region is required: all individual edge parameters remain in an open neighbourhood of the equal-pair section. Compose these sections through the finite cleanup sequence; disjoint effective edges have their own adjustable coordinates. Dangling edges can be given arbitrary strictly interior parameters.

Write the lifted point as $\widetilde{\theta}_0$. With δ the weight of the retained incoming edge, the coordinates are $q = \delta q_{N-c} + (1 - \delta) q_{N-c'}$. At $(\widetilde{\theta}_0, 1)$ the residual map $F = (E_1, E_2, E_3)$ vanishes and has rank-three derivative in the old-edge directions because the cleanup map is a submersion. The formula extends

smoothly across this mixing-weight boundary. Select a nonsingular three-column minor and apply the implicit function theorem, fixing the other coordinates. It supplies solutions for $\delta < 1$ sufficiently near one. Strict admissibility of every edge, of the reconstructed tree (a continuous function of the positive coordinates), and nonvanishing of the minor persist by openness. □

Lemma 5 needs some N to apply to. Given a reduced trinet with a non-trivial three-blob, subdivide an edge of the blob by a new vertex s, subdivide the pendant edge of a leaf ℓ by a new vertex v, and add the edge $s \to v$. The result is a valid semi-directed network, since v's only descendant is ℓ. It is again reduced, by a short case analysis on whether s and v belong to a putative 2-sub-blob (Appendix B), and its blob has one more reticulation. Deleting $s \to v$ returns the original. Iterating from L2-1, and then attaching pendant subtrees, gives the main consequence.

Corollary 6. *For every $n \geq 3$ and every $k \geq 2$ there is an n-leaf network whose only non-trivial blob is a reduced three-blob of level k, and whose K2P model on Θ_{tb} meets the model of an n-leaf binary tree. Because a K2P edge with parameters (X, Y) is a K3P edge with Fourier parameters (X, Y, X), the same holds for the K3P model on its space Θ_0. Consequently the tree–network distinguishability theorem of Brits et al. (their Corollary 5.7) does not extend to K2P or K3P for any number of leaves.*

Numerical continuation along Lemma 5 confirms the construction at levels 3 and 4. Starting from Theorem 1, with the new mixing weight lowered to 0.9, it reaches coincidence points at which the normalised binomials $\log(A_i^2 B_j B_k) - \log(B_i C_i^2)$ are below 10^{-14}, with every edge strictly inside Θ_{tb} and all mixing weights between 0.49 and 0.93.

Lemma 5 also decides every level-3 trinet. Of the 83 reduced level-3 trinets, 30 are anchored and are therefore distinguishable by Theorem 3. Each of the other 53 has a reticulation edge whose deletion leaves, without any 2-sub-blob suppression, a copy of L2-0 (26 cases), L2-2 (21) or L2-1 (6), counting the first such edge found. The certificates of Theorems 1 and 2 are regular points strictly inside Θ_0, with exactly computed Jacobian rank 3, so each of these 53 coincides with a tree on Θ_0.

Theorem 7 (classification up to level three; computer-assisted). *A reduced trinet of level at most three with a non-trivial three-blob is distinguishable from star trees on Θ_0 if and only if it is anchored at some leaf.*

The enumerations behind Theorem 7 are computational, and the code is part of the accompanying material. We have not enumerated level 4 exhaustively. Inserting one new reticulation edge, in every acyclic way, into a rooted representative of each level-3 trinet produced 1,614 distinct reduced level-4 trinets. Of these, 443 are anchored and so distinguishable. Each of the remaining 1,171 has a single deletion leading to a non-anchored level-3 trinet, so by Lemma 5 each coincides with a tree.

Conjecture 2. *For every level, a reduced trinet with a non-trivial three-blob is distinguishable from star trees on Θ_0 under K2P if and only if it is anchored at some leaf.*

The anchored counts are 1 of 1, 3 of 6 and 30 of 83 in the exhaustive classifications through level three. The 443 of 1,614 figure at level four describes only the specified insertion-generated family; it is neither an exhaustive count nor an unbiased estimate of the fraction among all level-four trinets. Level-four and continuation outputs remain exploratory historical evidence, not part of the exact finite-classification gate.

On Θ_{emb} the picture is less complete. Fourteen level-3 trinets contain a copy of L2-1 in the sense of

Lemma 5, so they coincide with trees even on Θ_{tb}. One further level-3 trinet does not contain L2-1 but still appears to coincide on Θ_{emb}. In it, all three reticulations lie above leaves and share parents in a hexagon around a central tree vertex (index 5 in the data files). A Newton refinement carried to 50 significant digits reduces $|E|$ to 10^{-57}, with every inequality satisfied by a margin of at least 0.027. This is numerical evidence, not a certificate.

9. Why JC and K2P behave differently

Lemma 2 shows what is at stake. The star tree's G-parameters y_i are fixed by the pairwise coordinates B_i. Coincidence then requires each y_i to equal a weighted mean of the path values Y_i^T over displayed trees, with weights proportional to $w_T X_j^T X_k^T$.

Under JC every X_e equals Y_e, so the weights are made from the same numbers as the path values and the pairwise coordinates, and Brits et al.'s inequality shows that the mean always falls short. Under K2P the X_e are free, so the weights can be tuned independently of the Y values. Coincidence requires two things: each y_i must lie strictly between the smallest and largest path value Y_i^T, and the three sets of weights, which share the same X_e, must be tunable at the same time.

Anchoring describes when tuning is impossible. For the reticulation leaf x of a 3-sunlet, both displayed trees give the same product $X_j^T X_k^T$, namely the parameter of the cycle's tree edge. The weights are then exactly the mixing weights $(\delta, 1-\delta)$, and the JC inequality survives. In L2-1, by contrast, leaf 1 sees the four displayed trees with $X_2^T X_3^T \in \{X_1X_3, X_2X_4\}$, two products whose ratio is free. In Theorem 1 the weights for leaf 1 are proportional to $(51, 1, 51, 1)$, and the weighted mean of the path values $(464, 232, 684, 760)/1145$ is exactly $y_1 = 1/2$. Leaves 2 and 3 are handled in the same way, and the remaining coupling between the three weightings is one quadratic equation.

10. Open questions

1. *Continuous-time parameters.* Prove or refute Conjecture 1, and classify trinets on Θ_{emb}; the level-3 example in Section 8 suggests that containing L2-1 is not the only route to coincidence there.
2. *All levels.* Prove Conjecture 2. Theorem 3 gives one direction. For the other, it would suffice to show that every non-anchored trinet of level $k \geq 3$ has a single reticulation-edge deletion leading, without 2-sub-blob suppression, to a non-anchored trinet of level $k-1$. This holds for all 83 level-3 trinets and all 1,614 level-4 trinets generated here.
3. *Shared rate ratios.* Three questions differ: both models share the same prescribed ratio; each uses its own constant ratio; or only the network has a constant ratio and the tree is edge-specific. Proposition 8 below excludes the first for L2-1 when $\kappa > 1$. The second remains open. The historical `homogeneous_kappa.py` search constrained only the network and did not check tree admissibility, so its $\kappa = 10$ residual hit is not an admissible certificate. The revised script checks reconstructed-tree admissibility and labels any output as numerical, network-only evidence. We do not rely on the historical hit.
4. *K3P.* Since L2-3's model is contained in that of a 3-sunlet, the K3P level-1 result of Brits et al. (their Lemma 4.2) makes L2-3 distinguishable under K3P. The K3P status of L2-4 and L2-5

is open, because no single K3P binomial plays the role of E_x.

5. *Level-2 identifiability.* Englander et al. (2025, bioRxiv version 1) establish JC identifiability under additional structural hypotheses. We do not transfer its theorem numbering across versions or infer a failure of generic identifiability from an exceptional coincidence. Theorem 2 blocks an unrestricted K2P trinet-separation argument on Θ_0. Restrictions on parameters or additional four-leaf information remain possible routes. Holtgrefe et al. (2025) study quartet and inter-taxon quartet-distance information under specified structural assumptions; their data representation and question differ from the full-distribution coincidence here.

Proposition 8 (a common prescribed ratio). *L2-1 cannot coincide with a tree if every edge of both models has the same fixed transition/transversion ratio $\kappa > 1$, positive branch lengths and strictly positive mixing weights.*

Proof. Put $r = (\kappa + 1)/2 > 1$ and $u_e = Y_e$, so $X_e = Y_e^r$. For each leaf pair (j, k) let $z_T = Y_j^T Y_k^T$. Lemma 2 gives $A_i = \sum_T w_T z_T^r$ and $B_i = \sum_T w_T z_T$. A tree with this ratio requires $A_i = B_i^r$. Strict convexity and positive weights force all z_T equal for each pair. Removing common pendant factors, the four path triples are

$$(u_0, u_1, u_3), \quad (u_0u_3, u_2u_4, 1), \quad (u_8u_4, u_1u_3, 1), \quad (u_8, u_2, u_4).$$

The pair $(2, 3)$ yields $u_1u_3 = u_2u_4$, whereas the pair $(1, 2)$ from the first two triples yields $u_1 = u_2u_3u_4$. Since parameters are positive, substitution gives $u_3^2 = 1$, contradicting $0 < u_3 < 1$. The case $\kappa = 1$ is JC and is already excluded by the source theorem. Nothing here excludes two different globally constant ratios or an edge-specific comparison tree. □

11. Data and code

The release command `python code/verify_release.py --out NEW_DIRECTORY` runs exact probability comparison, derivative checks, class/isomorphism and deletion-route checks, and semantic negative controls. Running it with `python -O` retains the explicit scientific failure checks. It writes only to the requested new output directory. Historical searches are classified separately in `STATUS.json`; small residuals, finite-grid searches and the generated level-four family are not promoted to exact certificates.

For each seed, columns (X_0, X_1, X_3) give the following exact nonzero minors of $D(E_1, E_2, E_3)$:

$$\begin{aligned}
\text{L2–}\,0: &\quad \frac{2595026403}{33416413184000000000000000}, \\
\text{L2–}\,1: &\quad \frac{4090396401}{15233183847022592000000000000}, \\
\text{L2–}\,2: &\quad \frac{331012580224672537299106333883811018654233924900 7963}{11598918790923679142622803037741745115590819840000000000000000000000000}.
\end{aligned}$$

The retained machine receipt also gives a nonzero nine-column minor of the nine-coordinate map at each seed. These finite exact computations support regularity; they are not formal verification of the universal propagation argument.

The accompanying bundle contains the exact certificates (`certificates/*.json`, together with the exact network distributions in `*_result.json`) and the probability-space verifier `certify.py`. It also contains the enumeration code (`trinets.py`, `enumerate_level3.py`), the symbolic K2P model (`k2p_model.py`), the anchoring test (`anchoring.py`), the level-3 and level-4 analyses, the searches and reductions referred to above, and logs of every run. The certificates use exact rational arithmetic, with arithmetic in $\mathbb{Q}(\sqrt{P})$ for the star-tree parameters.

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Appendix A. Certificates for L2-0 and L2-2

Edges are listed as in Figure 1. Pendant edges have $(X, Y) = (1/2, 3/4)$ in both certificates. All parameters lie strictly inside Θ_0, and the exact Jacobian of (E_1, E_2, E_3) has rank 3 at both points.

L2-0 (reticulation h_0 with parents t_0, t_2 and weight $4/5$ on $t_0 \to h_0$; reticulation h_1 with parents t_0, t_1 and weight $1/2$ on $t_0 \to h_1$):

Edge	X	Y
$t_0 \to h_0$	16/45	20/43
$t_0 \to h_1$	10/21	1/20
$t_1 \to h_1$	1/2	79/100
$t_1 — t_0$	21/44	1/5
$t_2 — t_1$	15/28	1/10
$t_2 \to h_0$	1/2	7/43

The induced star tree has $y = (3/20, 3/20, 3/20)$ and $x \approx (0.1639, 0.1848, 0.2635)$, all in Θ_0.

L2-2 (reticulation h_0 with parents t_0, t_2 and weight $2/5$ on $t_0 \to h_0$; reticulation h_1 with parents t_0, t_1 and weight $2/5$ on $t_0 \to h_1$):

Edge	X	Y
$t_0 \to h_0$	1572832833/1853838968	3/4
$t_0 \to h_1$	1/5	80/141
$t_1 \to h_1$	36588927/73906976	305/1692
$t_2 — t_0$	2197123/4619186	1/20
$h_0 — t_1$	3/4	3/4
$t_2 \to h_0$	837103863/1853838968	11/72

The induced star tree has $y = (3/20, 3/5, 3/40)$ and $x \approx (0.1835, 0.4429, 0.1830)$, all in Θ_0.

Appendix B. The one-arc extension keeps a trinet reduced

Let N be a reduced trinet whose internal vertices form a single three-blob, and write ∂U for the number of edges joining a set U of vertices to its complement.

Fact. Every non-empty proper subset U of the internal vertices of N has $\partial U \geq 3$. If $\partial U \leq 1$, the joining edge (if any) is a bridge, hence a pendant edge $u\ell$, and then the other two leaves are cut off from U. If $\partial U = 2$ and U induces a connected subgraph, U is a 2-sub-blob or a single vertex, which has degree three. If $\partial U = 2$ and U is disconnected, one of its components has $\partial \leq 1$.

Let N^+ be obtained by subdividing a blob edge $\{p, q\}$ with a vertex s and the pendant edge $\{u, \ell\}$ with a vertex v, and adding the arc $s \to v$. Suppose W is a set of at least two internal vertices of N^+ that induces a connected subgraph and has $\partial W = 2$ in N^+. In each case below we find a non-empty proper subset of the internal vertices of N with $\partial \leq 2$ in N, contradicting the Fact.

- *Neither s nor v in W.* If $p, q \in W$, the edges ps and sq are the two leaving edges and $\partial_N W = 0$. Otherwise the leaving edges of W in N^+ correspond one-to-one with those in N, so $\partial_N W = 2$. In both cases W is proper, since the set of all internal vertices of N has five leaving edges in N^+.
- *$v \in W$, $s \notin W$.* The edges $v\ell$ and vs leave W, so $u \in W$ and nothing else leaves. Then $W \setminus \{v\}$ has $\partial_N = 1$.
- *$s, v \in W$.* The edge $v\ell$ leaves W, and at most one further edge does. If $u \notin W$, that edge is vu, so $p, q \in W$ and $\partial_N(W \setminus \{s, v\}) = 0$. If $u \in W$, then $\partial_N(W \setminus \{s, v\}) \leq 2$. The set $W \setminus \{s, v\}$ is non-empty, because $\{s, v\}$ alone has four leaving edges, and it is proper, because the set of all internal vertices of N^+ has three leaving edges.
- *$s \in W$, $v \notin W$.* The edge sv leaves W, and at most one further edge does, so $\partial_N(W \setminus \{s\}) \leq 1$. Here $W \setminus \{s\}$ is non-empty because $|W| \geq 2$.

Hence N^+ has no 2-sub-blob. The vertices s and v lie on a cycle through u and the old blob, so they join the blob, whose number of reticulations rises by one. An orientation of N extends to N^+ without directed cycles, because v's only descendant is ℓ.